

☐ أ٤

☐ ب٤

☐ ج٤

☐ د٤

☐ ه٤

6

* رقم الكمبيوتر

18

7

* الكود (قد تمت مراجعة بيانات الطالب ورقم الجلوس)

BOOD

 Will be reviewed

8

The Modelview matrix after executing the following OpenGL code segment is:

```
glMatrixMode(GL_MODELVIEW);
```

```
glLoadIdentity();
```

```
glScalef(-1.0, 2.0, -1.0);
```

```
glRotatef(90.0, 0, 0, 1.0); *
```

A)	B)	C)	D)
$\begin{bmatrix} 0 & -1 & 0 & 0 \\ -2 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 0 & -2 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 2 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

☐ A

☐ B

☐ C

☒ D

 **Will be reviewed**

9

Given three vertices $P1=(1, 1, 0.5)$, $P2=(1.5, 0.5, 0.5)$, $P3=(1, 0.5, 1)$ on a plane, the equation of the plane is *

☒ $x+y+z=2.5$

☐ $x+2y+2z=3.5$

☐ $x+2y+z=3$

☐ $x+y+0.5z=2.25$

 **Will be reviewed**

10

The position of the vertex after executing the following OpenGL code segment is:

```
glMatrixMode(GL_MODELVIEW);  
glLoadIdentity();  
glTranslatef(1.0f, 1.0f, 0.0f);  
glRotatef(90.0, 0, 0, 1.0);  
glScalef (-2.0,1.0, 1.0);  
glBegin(GL_POINTS);  
glVertex3f(1.0, 0.0, 0.0);  
glEnd(); *
```

☐ (0,1,0)

☐ (-1,0,0)

☐ (1,-1,0)

☒ (-1,-1,0)

✍ Will be reviewed

11

What is the primary function of glutPostRedisplay in OpenGL? *

☒ To request a redraw of the current window.

☐ To enter the main event processing loop.

☐ To initialize the OpenGL environment.

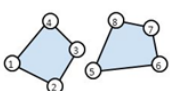
☐ To create a window on the screen.

✍ Will be reviewed

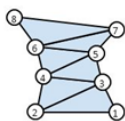
12

Select the output of executing the following OpenGL code segment:

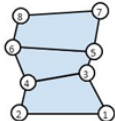
```
glBegin(GL_QUAD_STRIP);  
glVertex2fv(v1); glVertex2fv(v2);  
glVertex2fv(v3); glVertex2fv(v4);  
glVertex2fv(v5); glVertex2fv(v6);  
glVertex2fv(v7); glVertex2fv(v8);  
glEnd(); *
```



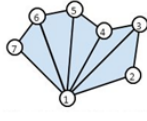
A



B



C



D

☐ A

☐ B

☒ C

☐ D

 **Will be reviewed**

13

Which matrix is used to project 3D objects onto a 2D screen? *

☒ Projection matrix

☐ Texture matrix

☐ Viewport matrix

☐ Modelview matrix

 **Will be reviewed**

14

In OpenGL, lines color is set using_____ *

☐ glColorF()

☒ glColor3f()

☐ glColor()

☐ glLineColor()

 **Will be reviewed**

15

How many parameters does glRectf take? *

☐ 2

☒ 4

☐ 8

☐ 6

 **Will be reviewed**

16

What is the purpose of the glPolygonMode function? *

☒ To set the mode in which polygons are rendered

☐ To set the depth test function

☐ To set the texture of a polygon

☐ To set the color of a polygon

 **Will be reviewed**

17

1. The Modelview matrix after executing the following OpenGL code segment is:

```
glMatrixMode(GL_MODELVIEW);  
glLoadIdentity();  
glRotatef(180.0, 0, 0, 1.0);  
glScalef(-1.0, 1.0, 1.0); *
```

A)	B)	C)	D)
$\begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

- ☐ A
- ☐ B
- ☒ C
- ☐ D

 **Will be reviewed**

18

What is the primary function of glVertex in OpenGL? *

- ☒ To define a vertex in 3D space.
- ☐ To draw a line between two points.
- ☐ To set the color of vertices.
- ☐ To set the color of subsequent vertices.

 **Will be reviewed**

19

What is the primary function of glutTimerFunc in OpenGL? *

- ☒ To register a timer callback function.
- ☐ To create a window on the screen.
- ☐ To enter the main event processing loop.

☐ To initialize the OpenGL environment.

 **Will be reviewed**

20

The 3x3 matrix for a translation by 2 in x and 3 in y is what? *

A. $\begin{pmatrix} 2 & 0 & 1 \\ 0 & 3 & 1 \\ 0 & 0 & 1 \end{pmatrix}$ B. $\begin{pmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ C. $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 3 & 1 \end{pmatrix}$ D. $\begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{pmatrix}$

☐ A

☐ B

☐ C

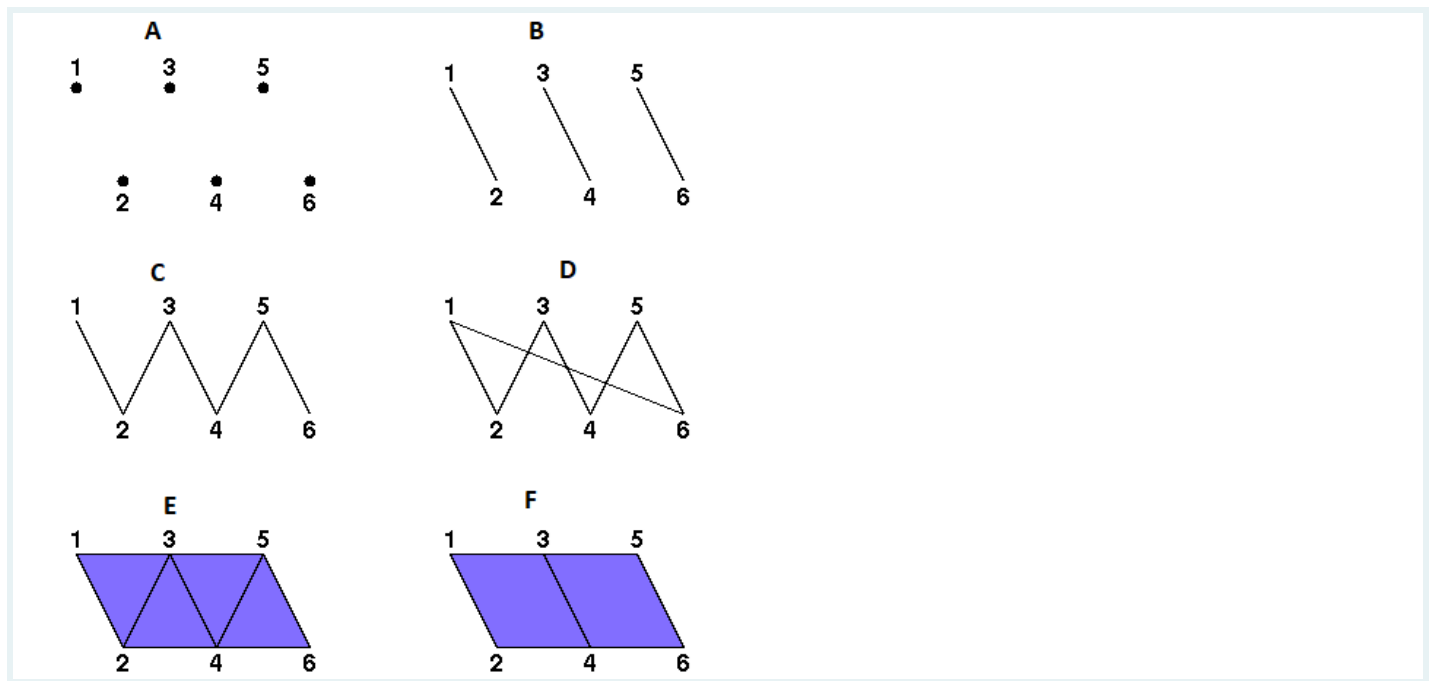
☒ D

 **Will be reviewed**

21

Select the output of executing the following OpenGL code segment:

```
glBegin(GL_LINE_STRIP);  
glVertex2fv(v1); glVertex2fv(v2);  
glVertex2fv(v3); glVertex2fv(v4);  
glVertex2fv(v5); glVertex2fv(v6);  
glEnd(); *
```



☐ A

☐ B

☒ C

☐ D

☐ E

☐ F

Will be reviewed

22

What is the relationship between `glClear` and `glClearColor`? *

☐ `glClear` sets the color to clear the color buffer, while `glClearColor` actually clears the buffer.

☐ There is no direct relationship between `glClear` and `glClearColor`.

☐ Both functions are used to clear the color buffer, but with different parameters.

- ☒ glColor sets the color to clear the color buffer, while glClear actually clears the buffer.

 **Will be reviewed**

23

How does glColor affect the appearance of primitives drawn after it is called? *

- ☐ It only affects the color of the next primitive.
- ☐ It only affects the color of the preceding primitive.
- ☐ It affects the color of all primitives in the scene.
- ☒ It sets the color for all subsequent primitives until a new glColor call is made.

 **Will be reviewed**

24

To draw a rectangle with upper-left corner (10, 20) and lower-right corner (30, 40), how would you call glRectf? *

- ☒ glRectf(10, 20, 30, 40);
- ☐ glRectf(30, 40, 10, 20);
- ☐ glRectf(20, 10, 40, 30);
- ☐ glRectf(40, 30, 20, 10);

 **Will be reviewed**

25

When is the glutReshapeFunc callback function invoked? *

- ☐ When a key is pressed.
- ☐ When mouse is moved.
- ☒ When the window is resized.
- ☐ When the mouse is clicked.

 **Will be reviewed**

26

Rotating the 3D point (0,-1,0) by -90 around z-axis gives what: *

- ☒ (- 1 , 0 , 0)
- ☐ (0 , -1 , 0)
- ☐ (1,0,0)
- ☐ (0 , 1 , 0)

 **Will be reviewed**

27

What happens inside the glutMainLoop() function? *

- ☐ It continuously checks for events like keyboard input, mouse input, and window resizing.
- ☐ It calls the registered display callback function to redraw the scene.
- ☐ It handles window management tasks like creating, destroying, and resizing windows.
- ☒ All of the above.



This content is created by the owner of the form. The data you submit will be sent to the form owner. Microsoft is not responsible for the privacy or security practices of its customers, including those of this form owner. Never give out your password.

Microsoft Forms | AI-Powered surveys, quizzes and polls [Create my own form](#)

[Privacy and cookies](#) | [Terms of use](#)